

Software Requirements Specification

Mboa Private Language Workspace

Hosted Business Release

Engineering team

Kwete Ngouba Junior Rayan

Djemtchimo Noukui Bruno Jonatan

Amina Boubakary

Version 2.0 — September 20, 2026

Private workspaces. Explicit processing. Traceable review.

Release boundary. This specification covers implemented business software and its acceptance requirements. Automated fixtures do not establish live-provider quality, infrastructure readiness or launch approval.

Approval owner: _____

Approval date: _____

Contents

1	Purpose and Document Control	3
1.1	Status vocabulary	3
2	Product and Operating Scope	3
2.1	Users and services	3
2.2	Business areas	3
2.3	Languages and evidence	4
3	Functional Requirements	4
4	Data and Retention Requirements	6
4.1	Authoritative storage	6
4.2	Explicit bounds	6
4.3	Retention and deletion	7
5	Non-functional Requirements	7
6	Acceptance and Release Gates	8
6.1	Automated completion	8
6.2	Owner-supplied acceptance	8
6.3	Scope not silently added	9
7	Traceability	9

1 Purpose and Document Control

This SRS defines a web application for individual users who need translation, terminology, document/audio processing and recoverable saved work. Each account has its own private workspace; projects organize one owner's work. Local testing exercises the same authenticated application intended for hosting.

Version 2.0 supersedes the local-only operating assumptions of version 1.3. The earlier academic requirements and planning estimates remain in `docs/srs-legacy.tex`; they are not this release's business backlog, acceptance evidence or user interface. No historical sprint is retroactively declared completed. School identities and research records are not onboarding data for hosted users.

The SDD describes source-grounded implementation and UML. The current `README.md` describes installation and operation. Reproducible verification and remaining owner gates are recorded in `docs/README.md` and `docs/evidence`.

1.1 Status vocabulary

Implemented

Present in production source, with relevant automated checks. This does not certify every deployment or external service.

Automated evidence

A recorded test/build/measurement using stated fixtures and environment, not a claim about real users or live language quality.

Owner gate

Infrastructure, credentials, content rights, human evaluation or a policy decision requiring the release owner.

Excluded

No commitment to deliver that capability in this release.

2 Product and Operating Scope

2.1 Users and services

A visitor can register, verify an email, sign in or recover an account. A signed-in owner manages only that owner's data. There are no shared-team roles, cross-account projects, billing or in-app administrator console. The service operator configures infrastructure and can access the underlying storage; privacy is account isolation, not end-to-end encryption.

Supporting systems are SMTP for account emails, optionally Google for identity, and optionally Gemini for generation/transcription. Browser speech services have their own implementation-dependent processing boundary. The application does not promise native Francanglais or Cameroon Pidgin voices.

2.2 Business areas

The application has seven areas: Translate, Assistant, Terminology, Dictionary, Documents & audio, History, and Workspace settings. It does not expose the preserved compiler coursework pages or constructed-example library by default. The standalone desktop collector and legacy global CSV remain separate.

2.3 Languages and evidence

French, English, Francanglais/Camfranglais and Cameroon Pidgin are supported language directions, not interchangeable labels. The reference dictionary contains 179 source rows with English meanings, including repetitions and competing senses. It contains neither reviewed French glosses nor reliable grammatical-category annotations. Approval of private terminology records the owner's review, not expert certification or universal linguistic truth.

3 Functional Requirements

ID / area	Requirement and measurable acceptance
BR-01 Accounts	Register with a valid email, display name and password of 12–128 characters. Require email verification before password sign-in. Acceptance: An unverified login is refused; verification consumes a single-use, one-hour token. Registration never returns that token in the API response.
BR-02 Sessions	Use server-validated cookie sessions, explicit sign-out, origin checks and session-specific CSRF protection on authenticated mutations. Acceptance: Private APIs return 401 without a valid identity. Wrong Origin or CSRF returns 403. Production cookies are Secure, HttpOnly and SameSite=Lax.
BR-03 Recovery	Provide resend-verification and password-reset emails with generic public responses; revoke all prior sessions after a successful password reset. Acceptance: Expired/replayed tokens fail. Old credentials/sessions cannot access saved work after reset. Email-service failures are explicit and retryable.
BR-04 Google	Support Google authorization-code sign-in with PKCE, state, nonce and signed identity-token validation when operator credentials are configured. Acceptance: Wrong issuer/audience/expiry/nonce, replayed state or unverified email is rejected. Matching email does not automatically link an existing password account; explicit signed-in linking is required.
BR-05 Isolation	Choose private storage using the authenticated, server-generated account ID. Bind the selected project to the request, not process-global mutable state. Acceptance: Two accounts cannot read, change, restore or download one another's terms, saved work, media, backups or preview tokens. Parallel request tests remain isolated.
BR-06 Projects	Create and rename private projects; retain General; permit deletion only of empty projects without terminology, saved work or revisions. Acceptance: A foreign project is rejected. Switching warns before discarding drafts. The active project must be switched before its UI deletion; account-level management remains available if a formerly selected project is removed.

ID / area	Requirement and measurable acceptance
BR-07 Terms	Create, search, filter, edit and explicitly delete original expressions, language, glosses and business metadata. Preserve identifiers and provenance. Acceptance: Duplicates in the same language return 409. An evidence edit invalidates approval unless the exact new evidence is explicitly reviewed. Failed saves do not silently discard drafts or pretend success.
BR-08 Translation	Prefer unambiguous whole-entry approved terminology and enabled reference alignments. Label optional AI suggestions and their supplied evidence. Acceptance: Exact dictionary lookup works without Gemini. Local ambiguity or missing coverage with AI disabled returns an actionable error. Separate word matches are not represented as a trustworthy full-sentence translation.
BR-09 Assistant	Send an explicitly submitted message, bounded recent conversation and enabled source evidence to the configured provider. Acceptance: No request is sent on navigation or draft handoff. The provider never automatically approves or saves its output as private terminology.
BR-10 Dictionary	Search/paginate the supplied reference rows without mutating private terminology. Retain source document/line and competing senses. Acceptance: The 143 core and 36 supplementary rows remain separate references. Source errors are explicit, not an empty successful dictionary.
BR-11 Imports	Preview supported documents/media with bounded extraction. Require separate cloud consent and an explicit preview for processing that needs AI. Acceptance: Changing a file resets consent. Structured candidates remain unreviewed. Selecting, previewing or handing off a passage never automatically saves it.
BR-12 Recording	Record, stop, preview and download audio in supported browsers; upload only through explicit save or consented import actions. Acceptance: Permission denial and unexpected device stops are visible. Active recording prevents save; failed publication cleans only new uncommitted media. Typing remains available when recording is unavailable.
BR-13 Saved work	Explicitly save translations/conversations in the current project. Provide open, title search, rename, JSON download, draft reuse and deletion. Acceptance: Reload preserves saved items, not unsaved drafts. Account/project changes clear private transient UI and ignore late responses, including delayed bodies.
BR-14 Revisions	Record terminology create/update/delete/restore events transactionally and restore a selected revision using the current workspace version. Acceptance: A changed workspace causes 409 rather than overwrite. A recovered entry is unreviewed. A missing required recording or duplicate expression prevents restore.

ID / area	Requirement and measurable acceptance
BR-15 Backups	Create, list, download and deliberately delete verified full-account ZIPs. Include all projects, terms, revisions, saved work and managed audio. Acceptance: Manifest, paths, counts, schema, media signatures and referenced files are validated. Password/session databases and provider configuration are excluded. A failed backup is not published as usable.
BR-16 Restore	Preview an owner-scoped backup before replacement. Require literal REPLACE, an unexpired token and the unchanged workspace version. Acceptance: Create a verified safety copy first. Stage media without overwriting existing files; commit logical rows in one transaction. Reject stale/unsafe archives. Errors after possible commit explicitly advise inspecting saved state.
BR-17 Scheduling	Offer opt-in daily or weekly backups with 1–10 retained automatic copies. Acceptance: Check schedules each minute while the server runs. Preserve manual/safety copies. Record per-workspace failures and retry subsequent cycles after storage-scan failure; stopping the server stops scheduling.
BR-18 Operation	Serve the built website and API on one origin through a single launcher. Reject unsafe development exposure and incomplete production configuration. Acceptance: One command starts both surfaces. Production requires HTTPS origin and SMTP, forbids disabled auth/reload, requires built assets and disables API docs.
BR-19 Usage controls	Bound request bodies, retained data and provider attempts. Keep local lookup available when the daily AI allowance is exhausted. Acceptance: Every actual outbound attempt, including retry, consumes user and global allowance. Exact local results and missing-provider configuration consume none.

4 Data and Retention Requirements

4.1 Authoritative storage

The identity database contains accounts, hashed sessions, one-time tokens, OAuth state and rate counters. Each account's workspace database contains projects, entries, revisions, history and metadata. Audio and ZIPs live in that same account directory. Server-generated identifiers and validation prevent client-supplied account paths.

Private records use the same fourteen fields as the entry DTO. They include identifier, text, entry type, French and English glosses, category, source location, notes, audio filename, contributor, timestamp, language, review status and lexical category. SQLite is authoritative for hosted records; the desktop CSV is separate.

4.2 Explicit bounds

Resource	Limit
Projects per account	20, including General

Resource	Limit
Terminology / saved items / revisions	10,000 / 500 / 20,000 per account
Saved conversation	100 messages; each message at most 4,000 characters
Managed audio	1,000 files and 128 MiB aggregate per account
Single import/audio upload	12 MiB plus bounded multipart overhead
Document extraction	40 PDF pages; 12 megapixels per image; 40,000 characters; 100 aligned draft records
Translation input	4,000 characters
Backup archive	32 MiB compressed; 160 MiB expanded; 1,002 members
Backup storage / previews	512 MiB per account; at most three one-hour previews
General JSON request	1 MiB
AI daily attempts (defaults)	100 per account; 2,000 globally, including retries

Exceeding a bound is an explicit failure, not silent truncation of saved work. Limits are safeguards, not purchased storage entitlements or certified capacity.

4.3 Retention and deletion

Deleting a current record does not erase previous revisions or backup copies. Committed media is retained for deliberate recovery; imports do not retain raw uploaded files as a document library. A workspace restore replaces logical content but does not replace accounts, quotas or credentials.

The owner must approve retention, deletion requests, off-host copies, legal notices and support procedures before launch. There is no hidden assertion of regulatory compliance or automated account-erasure workflow. These decisions must precede any destructive retention automation.

5 Non-functional Requirements

ID	Requirement and acceptance boundary
NBR-01 Privacy	Authenticate and authorize at the API boundary, not only by hiding UI. Use bounded, parameterized storage operations and server-only secrets. Acceptance: Cross-account integration tests cover IDs, projects, media, history, archives and preview tokens. No real private dataset is used as a test fixture.
NBR-02 Reliability	Surface invalid input, provider errors, storage failure and uncertain commits. Use atomic publication and appropriate transaction/version checks. Acceptance: Injected failures preserve prior logical data and committed media as specified. Post-mutation refresh supersedes stale reads rather than dropping the refresh.

ID	Requirement and acceptance boundary
NBR-03 Response time	Move blocking private-store initialization and provider-accounting I/O off the ASGI event loop. Bound provider retries and UI request duration. Acceptance: A thread-identity regression verifies off-loop initialization. Record timings on the actual test host; do not claim an agreed public throughput/SLA without load targets, deployment measurement and operator approval.
NBR-04 Usability	Provide responsive desktop/mobile layouts, visible errors, explicit save/delete/restore actions, keyboard focus management and accessible form names. Acceptance: Desktop/mobile browser regressions exercise navigation, modal focus, review, recording permission errors and account/recovery workflows. Physical devices and assistive-technology acceptance remain human gates.
NBR-05 Code quality	Keep typed frontend sources, reusable request/storage helpers, reproducible builds, automated regressions and source-grounded documentation. Acceptance: Type-check app and tests; build assets; verify class coverage, sequence activations, atlas references and embedded PDF figure pixels.
NBR-06 Deployment	Run on one durable host with restricted storage and an HTTPS reverse proxy. Treat infrastructure monitoring and disaster recovery as explicit operations. Acceptance: Validate production settings and headers in tests. Owner rehearses actual host restart, email delivery, TLS, disk recovery and off-host restore. No ephemeral-disk or multi-host clustering support is implied.

6 Acceptance and Release Gates

6.1 Automated completion

The release verification register must contain observed results, exact commands, environment and limitations. Required checks cover Python domain/API/storage regressions, frontend unit/component behavior, app/test TypeScript, production build, mocked desktop/mobile scenarios and real local full-stack browser flows. Local fake email, signed mock Google tokens, generated audio and mocked Gemini responses must never be described as live-provider acceptance.

CI retains Windows Python 3.11 and 3.14, adds Linux server checks and real full-stack browser flows, and verifies documentation. A local working-tree run does not establish remote CI success; publication and a passing workflow remain an explicit release gate.

6.2 Owner-supplied acceptance

1. Hosting account, public domain, TLS, durable disk, trusted-proxy settings, service supervision, monitoring and off-host recovery destination.
2. Configure the SMTP sender and credentials, then verify actual delivery of account messages. Configure the Google client, consent screen and authorized callbacks.

3. Gemini access/budget and authorized non-sensitive live trials, with human review of translation/transcription quality and actual service errors.
4. Approved business glossary, reviewed French meanings and source-content redistribution permissions.
5. Physical microphone/speaker, denied permissions, device disconnect, browser-voice, real-device and assistive-technology acceptance.
6. Privacy/retention/terms/support, capacity and recovery objectives, change publication and final launch approval.

6.3 Scope not silently added

Billing, subscriptions, shared-organization permissions, fine-tuned language models, certified legal/medical translation and distributed multi-host storage are excluded. The earlier 8 GB reference-machine desktop performance target is not met merely by measurements on a roughly 32 GB development machine. It is retained as historical context, not an undisclosed hosted SLA.

7 Traceability

Requirements	Implementation and verification
BR-01–05	<code>backend/auth.py</code> ; hosted integration tests; AuthGate unit tests; real browser registration/recovery/isolation
BR-05–07, 13–14	<code>backend/workspaces.py</code> ; dataset adapter; collection/audio APIs; workspace/settings/history components
BR-08–11	main/grounding/dictionary/import modules; Gemini regressions; Translator/Imports unit and browser tests
BR-12	browser recorder and audio API regressions; synthetic audio browser stream; physical acceptance separately
BR-15–17	<code>backend/workspace_backups.py</code> ; corruption/conflict/rollback/maintenance tests; full-stack restore
BR-18–19	<code>tools/run_app.py</code> , <code>backend/web.py</code> ; launcher/body/header/provider-accounting tests
NBR-01–06	full verification register, UML/PDF checker and owner acceptance record